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A Trivially Short SAND Report (justified, report-style)

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ABSTRACT

So abstract it hurts.

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1. INTRODUCTION

Very glad to meet you.

1.1. A Section

Underpromising, and hopefully over-delivering.

1.2. Another Section

Look at us go!

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2. BACKGROUND

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3. OBJECTIVE

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4. DATA

4.1. C-Frame

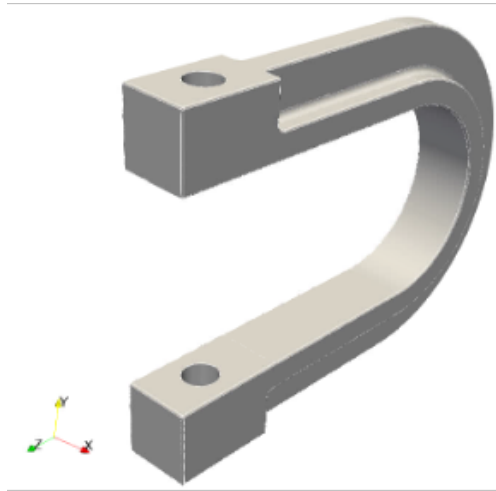


Figure 4-1. C-frame geometry used for modal analysis.

- Analysis type: modal
- Material: isotropic linear elastic
- Mesh: IGA
- Geometry: C-frame
- Boundary conditions: fixed at base
- Loads: N/A
- Solver: direct

4.1.1. cframe_modal/jobs_coarse.iga

```
1 {  
2   "coreform_iga_version": "2025.7",  
3   "data_output_file": "cf_iga_data_output.json",  
4   "data_output_interval": "intervals_1",  
5   "flex_models": [  
6     {  
7       "database_name": "geom",  
8       "label": "flex_models_1",
```

```

9     "parts": [
10         {
11             "label": "parts_1",
12             "material": "materials_1",
13             "material_model": "elastic",
14             "part": "part_1",
15             "quadrature_options": {
16                 "optimization_lawson_hanson_solve_completion": true,
17                 "optimization_method": "tet_pruning",
18                 "optimization_residual_tolerance": 1.0000000000000002e-12,
19                 "optimized_cut_quadrature_strategy": "reduce_from_uncut",
20                 "quadrature": "QP1"
21             }
22         },
23     ],
24     "small_cell_volume_ratio": 0
25 },
26 ],
27 "functions": [
28     {
29         "constant": {
30             "value": 1
31         },
32         "label": "functions_1"
33     }
34 ],
35 "intervals": [
36     {
37         "label": "intervals_1",
38         "start_step": 0,
39         "step_increment": 1,
40         "stop_step": 20,
41         "time_increment": 1
42     }
43 ],
44 "label": "iga_1",
45 "materials": [
46     {
47         "elastic": {
48             "poissons_ratio": 0.29,
49             "youngs_modulus": 280000000
50         },
51         "label": "materials_1",
52         "mass_density": 0.000735
53     }
54 ],
55 "procedures": [
56     {
57         "label": "procedures_1",
58         "structural_modal_analysis": {
59             "boundary_conditions": [
60                 "boundary_conditions_1"
61             ],
62             "flex_model": "flex_models_1",
63             "modal_solution_method": "modal_solution_methods_1",
64             "outputs": [
65                 "outputs_1"
66             ]
67         }
68     }
69 ],
70 "structural_modal_analysis_definitions": {
71     "boundary_conditions": [
72         {
73             "components": [
74                 "x",
75                 "y",
76                 "z"
77             ],
78             "label": "boundary_conditions_1",
79             "penalty": {
80                 "dimensionless": true,
81                 "value": 10
82             },
83             "quadrature_options": {
84                 "optimization_lawson_hanson_solve_completion": true,
85                 "optimization_method": "tet_pruning",
86                 "optimization_residual_tolerance": 1.0000000000000002e-12,
87                 "optimized_cut_quadrature_strategy": "reduce_from_uncut",
88                 "quadrature": "QP1"
89             },
90             "set": "fixed"
91         }
92     ],
93     "modal_solution_methods": [
94         {
95             "frequency_extraction": {
96                 "maximum_frequency": 4000,
97                 "minimum_frequency": 1
98             },
99             "label": "modal_solution_methods_1"
100         }

```

```

101 ],
102 "outputs": [
103   {
104     "field": {
105       "database_name": "cframe_modal_coarse",
106       "external_surfaces_only": false,
107       "include_integration_point_output": false,
108       "interval": "intervals_1",
109       "part": "part_1",
110       "variables": {
111         "displacement": [
112           "all"
113         ]
114       }
115     },
116     "label": "outputs_1",
117     "tessellate": true
118   }
119 ]
120 }
121 }

```

4.1.2. cframe_modal/jobs_fine.iga

```

1 {
2   "coreform_iga_version": "2025.7",
3   "data_output_file": "cf_iga_data_output.json",
4   "data_output_interval": "intervals_1",
5   "flex_models": [
6     {
7       "database_name": "geom",
8       "label": "flex_models_1",
9       "parts": [
10        {
11          "label": "parts_1",
12          "material": "materials_1",
13          "material_model": "elastic",
14          "part": "part_1",
15          "quadrature_options": {
16            "optimization_lawson_hanson_solve_completion": true,
17            "optimization_method": "tet_pruning",
18            "optimization_residual_tolerance": 1.0000000000000002e-12,
19            "optimized_cut_quadrature_strategy": "reduce_from_uncut",
20            "quadrature": "QP1"
21          }
22        }
23      ],
24      "small_cell_volume_ratio": 0
25    }
26  ],
27  "functions": [
28    {
29      "constant": {
30        "value": 1
31      },
32      "label": "functions_1"
33    }
34  ],
35  "intervals": [
36    {
37      "label": "intervals_1",
38      "start_step": 0,
39      "step_increment": 1,
40      "stop_step": 20,
41      "time_increment": 1
42    }
43  ],
44  "label": "iga_1",
45  "materials": [
46    {
47      "elastic": {
48        "poissons_ratio": 0.29,
49        "youngs_modulus": 28000000
50      },
51      "label": "materials_1",
52      "mass_density": 0.000735
53    }
54  ],
55  "procedures": [
56    {
57      "label": "procedures_1",
58      "structural_modal_analysis": {
59        "boundary_conditions": [
60          "boundary_conditions_1"

```

```

61 ],
62 "flex_model": "flex_models_1",
63 "modal_solution_method": "modal_solution_methods_1",
64 "outputs": [
65   "outputs_1"
66 ]
67 }
68 },
69 ],
70 "structural_modal_analysis_definitions": {
71   "boundary_conditions": [
72     {
73       "components": [
74         "x",
75         "y",
76         "z"
77       ],
78       "label": "boundary_conditions_1",
79       "penalty": {
80         "dimensionless": true,
81         "value": 10
82       },
83       "quadrature_options": {
84         "optimization_lawson_hanson_solve_completion": true,
85         "optimization_method": "tet_pruning",
86         "optimization_residual_tolerance": 1.0000000000000002e-12,
87         "optimized_cut_quadrature_strategy": "reduce_from_uncut",
88         "quadrature": "QP1"
89       },
90       "set": "fixed"
91     },
92   ],
93   "modal_solution_methods": [
94     {
95       "frequency_extraction": {
96         "maximum_frequency": 4000,
97         "minimum_frequency": 1
98       },
99       "label": "modal_solution_methods_1"
100     }
101   ],
102   "outputs": [
103     {
104       "field": {
105         "database_name": "cframe_modal_fine",
106         "external_surfaces_only": false,
107         "include_integration_point_output": false,
108         "interval": "intervals_1",
109         "part": "part_1",
110         "variables": {
111           "displacement": [
112             "all"
113           ]
114         }
115       },
116       "label": "outputs_1",
117       "tessellate": true
118     }
119   ]
120 }
121 }

```

4.1.3. cframe_modal/jobs_fine_x2.iga

```

1 {
2   "coreform_iga_version": "2025.7",
3   "data_output_file": "cf_iga_data_output.json",
4   "data_output_interval": "intervals_1",
5   "flex_models": [
6     {
7       "database_name": "geom",
8       "label": "flex_models_1",
9       "parts": [
10        {
11          "label": "parts_1",
12          "material": "materials_1",
13          "material_model": "elastic",
14          "part": "part_1",
15          "quadrature_options": {
16            "optimization_lawson_hanson_solve_completion": true,
17            "optimization_method": "tet_pruning",
18            "optimization_residual_tolerance": 1.0000000000000002e-12,
19            "optimized_cut_quadrature_strategy": "reduce_from_uncut",
20            "quadrature": "QP1"

```

```

21     }
22   }
23 ],
24   "small_cell_volume_ratio": 0
25 }
26 ],
27 "functions": [
28   {
29     "constant": {
30       "value": 1
31     },
32     "label": "functions_1"
33   }
34 ],
35 "intervals": [
36   {
37     "label": "intervals_1",
38     "start_step": 0,
39     "step_increment": 1,
40     "stop_step": 20,
41     "time_increment": 1
42   }
43 ],
44 "label": "iga_1",
45 "materials": [
46   {
47     "elastic": {
48       "poissons_ratio": 0.29,
49       "youngs_modulus": 280000000
50     },
51     "label": "materials_1",
52     "mass_density": 0.000735
53   }
54 ],
55 "procedures": [
56   {
57     "label": "procedures_1",
58     "structural_modal_analysis": {
59       "boundary_conditions": [
60         "boundary_conditions_1"
61       ],
62       "flex_model": "flex_models_1",
63       "modal_solution_method": "modal_solution_methods_1",
64       "outputs": [
65         "outputs_1"
66       ]
67     }
68   }
69 ],
70 "structural_modal_analysis_definitions": {
71   "boundary_conditions": [
72     {
73       "components": [
74         "x",
75         "y",
76         "z"
77       ],
78       "label": "boundary_conditions_1",
79       "penalty": {
80         "dimensionless": true,
81         "value": 10
82       },
83       "quadrature_options": {
84         "optimization_lawson_hanson_solve_completion": true,
85         "optimization_method": "tet_pruning",
86         "optimization_residual_tolerance": 1.0000000000000002e-12,
87         "optimized_cut_quadrature_strategy": "reduce_from_uncut",
88         "quadrature": "QP1"
89       },
90       "set": "fixed"
91     }
92   ],
93   "modal_solution_methods": [
94     {
95       "frequency_extraction": {
96         "maximum_frequency": 4000,
97         "minimum_frequency": 1
98       },
99       "label": "modal_solution_methods_1"
100     }
101   ],
102   "outputs": [
103     {
104       "field": {
105         "database_name": "cframe_modal_fine_x2",
106         "external_surfaces_only": false,
107         "include_integration_point_output": false,
108         "interval": "intervals_1",
109         "part": "part_1",
110         "variables": {
111           "displacement": [
112             "all"

```

```

113     ]
114   },
115   },
116   "label": "outputs_1",
117   "tessellate": true
118 }
119 ]
120 }
121 }

```

4.2. Door Hinge

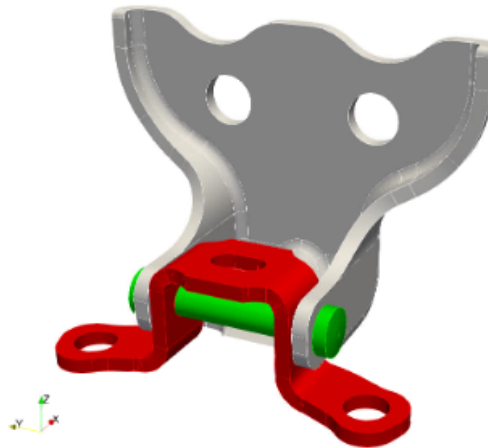


Figure 4-2. Door hinge geometry used for nonlinear static analysis.

4.2.1. door_hinge/door_hinge_static/door_hinge_failure_workflow.txt

```

1  #=Door Lock Hinge Assembly Problem
2  reset
3  dev rimport "/tutorials/door_lock_assembly/door_lock_assembly.sat"
4  #!=Door Lock Hinge Assembly Problem
5  #=DEFINE PARTS AND SETS
6  name part 1 "door_link"
7  name part 2 "door_hinge"
8  name part 3 "door_pin"
9  set "door_link_bottom_surfaces" add surfaces [21 29]
10 set "door_hinge_bolt_surfaces" add surfaces [111 115]
11 #!=DEFINE PARTS AND SETS
12 #=MESH GENERATION
13 mesh "mesh_door_link" rectilinear degree 2 continuity 1 element_size [1 1 1] padding [2 2 2]
14 part "door_link" mesh "mesh_door_link"
15 part "door_link" volume_box
16 mesh "mesh_door_hinge" rectilinear degree 2 continuity 1 element_size [1 1 1] padding [2 2 2]
17 part "door_hinge" mesh "mesh_door_hinge"
18 part "door_hinge" volume_box
19 mesh "mesh_door_pin" rectilinear degree 2 continuity 1 element_size [1 1 1] padding [2 2 2]
20 part "door_pin" mesh "mesh_door_pin"
21 part "door_pin" volume_box
22 #!=MESH GENERATION
23 #=FUNCTIONS
24 functions "aluminum_plastic_hardening" new
25 functions "aluminum_plastic_hardening" piecewise_linear abscissa [0 10]
26 functions "aluminum_plastic_hardening" piecewise_linear ordinate [0 100]
27 functions "steel_plastic_hardening" new
28 functions "steel_plastic_hardening" piecewise_linear abscissa [0 10]
29 functions "steel_plastic_hardening" piecewise_linear ordinate [0 100]
30 functions "constant_1" new
31 functions "constant_1" constant value 1
32 functions "linear_ramp" new

```

```

33 functions "linear_ramp" piecewise_linear abscissa [0 1]
34 functions "linear_ramp" piecewise_linear ordinate [0 1]
35 #!=FUNCTIONS
36 #=MATERIALS
37 materials "aluminum" new
38 materials "aluminum" mass_density 2.7e-9
39 materials "aluminum" neohookean_isotropic_plasticity neohookean youngs_modulus 73.8e3
40 materials "aluminum" neohookean_isotropic_plasticity neohookean poissons_ratio 0.3
41 materials "aluminum" neohookean_isotropic_plasticity isotropic_plasticity yield_stress 322
42 materials "aluminum" neohookean_isotropic_plasticity isotropic_plasticity hardening_function "aluminum_plastic_hardening"
43 materials "steel" new
44 materials "steel" mass_density 2.7e-9
45 materials "steel" neohookean_isotropic_plasticity neohookean youngs_modulus 200e3
46 materials "steel" neohookean_isotropic_plasticity neohookean poissons_ratio 0.29
47 materials "steel" neohookean_isotropic_plasticity isotropic_plasticity yield_stress 2e3
48 materials "steel" neohookean_isotropic_plasticity isotropic_plasticity hardening_function "steel_plastic_hardening"
49 tessellation_options "tessellation_options_1" meshgms average_triangle_edge_subdivisions 4
50 #!=MATERIALS
51 #=FLEX MODELS
52 flex_models "flex_models_1" new
53 flex_models "flex_models_1" database_name "geom"
54 flex_models "flex_models_1" small_cell_volume_ratio 0
55 flex_models "flex_models_1" parts "door_hinge_instance" part "door_hinge"
56 flex_models "flex_models_1" parts "door_hinge_instance" material "aluminum"
57 flex_models "flex_models_1" parts "door_hinge_instance" material_model neohookean_isotropic_plasticity
58 flex_models "flex_models_1" parts "door_hinge_instance" quadrature_options tessellation_option "tessellation_options_1"
59 flex_models "flex_models_1" parts "door_link_instance" part "door_link"
60 flex_models "flex_models_1" parts "door_link_instance" material "aluminum"
61 flex_models "flex_models_1" parts "door_link_instance" material_model neohookean_isotropic_plasticity
62 flex_models "flex_models_1" parts "door_link_instance" quadrature_options tessellation_option "tessellation_options_1"
63 flex_models "flex_models_1" parts "door_pin_instance" part "door_pin"
64 flex_models "flex_models_1" parts "door_pin_instance" material "steel"
65 flex_models "flex_models_1" parts "door_pin_instance" material_model neohookean_isotropic_plasticity
66 flex_models "flex_models_1" parts "door_pin_instance" quadrature_options tessellation_option "tessellation_options_1"
67 #!=FLEX MODELS
68 #=INTERVALS
69 intervals "simulation_interval" new
70 intervals "simulation_interval" start_time 0
71 intervals "simulation_interval" stop_time 1
72 intervals "simulation_interval" time_increment 5e-4
73 intervals "simulation_output_interval" new
74 intervals "simulation_output_interval" use_start_stop_from_interval "simulation_interval"
75 intervals "simulation_output_interval" step_increment 1
76 #!=INTERVALS
77 #=LINEAR EQUATION SOLVERS
78 linear_equation_solvers "direct_lu" new
79 linear_equation_solvers "direct_lu" direct lu
80 linear_equation_solvers "direct_multi_frontal" new
81 linear_equation_solvers "direct_multi_frontal" direct multi_frontal
82 #!=LINEAR EQUATION SOLVERS
83 #=NONLINEAR EQUATION SOLVERS
84 nonlinear_equation_solvers "newton_raphson" new
85 nonlinear_equation_solvers "newton_raphson" newton linear_equation_solver "direct_multi_frontal"
86 nonlinear_equation_solvers "newton_raphson" newton maximum_iterations 20
87 nonlinear_equation_solvers "newton_raphson" newton target_relative_residual 1e-8
88 #!=NONLINEAR EQUATION SOLVERS
89 #=TIMESTEPPERS
90 time_steppers "implicit_dynamics_quasistatic" new
91 time_steppers "implicit_dynamics_quasistatic" generalized_alpha spectral_radius 0
92 time_steppers "implicit_dynamics_quasistatic" generalized_alpha predictor constant_velocity
93 time_steppers "implicit_dynamics_quasistatic" generalized_alpha nonlinear_equation_solver "newton_raphson"
94 time_steppers "implicit_dynamics_quasistatic" generalized_alpha adaptivity maximum_time_step 0.1
95 time_steppers "implicit_dynamics_quasistatic" generalized_alpha adaptivity minimum_time_step 1e-5
96 time_steppers "implicit_dynamics_quasistatic" generalized_alpha adaptivity decrease_factor 0.25
97 time_steppers "implicit_dynamics_quasistatic" generalized_alpha adaptivity increase_factor 1.5
98 #!=TIMESTEPPERS
99 #=BOUNDARY CONDITIONS
100 solid_mechanics_definitions boundary_conditions "hold_door_link_bottom" new
101 solid_mechanics_definitions boundary_conditions "hold_door_link_bottom" set "door_link_bottom_surfaces"
102 solid_mechanics_definitions boundary_conditions "hold_door_link_bottom" displacement function "constant_1"
103 solid_mechanics_definitions boundary_conditions "hold_door_link_bottom" displacement components 0 x
104 solid_mechanics_definitions boundary_conditions "hold_door_link_bottom" displacement components 1 y
105 solid_mechanics_definitions boundary_conditions "hold_door_link_bottom" displacement components 2 z
106 solid_mechanics_definitions boundary_conditions "hold_door_link_bottom" displacement scale_factor 0
107 solid_mechanics_definitions boundary_conditions "hold_door_link_bottom" quadrature_options tessellation_option "tessellation_options_1"
108 solid_mechanics_definitions boundary_conditions "pull_door_hinge_bolts" new
109 solid_mechanics_definitions boundary_conditions "pull_door_hinge_bolts" set "door_hinge_bolt_surfaces"
110 solid_mechanics_definitions boundary_conditions "pull_door_hinge_bolts" displacement function "linear_ramp"
111 solid_mechanics_definitions boundary_conditions "pull_door_hinge_bolts" displacement components 0 x
112 solid_mechanics_definitions boundary_conditions "pull_door_hinge_bolts" displacement components 1 y
113 solid_mechanics_definitions boundary_conditions "pull_door_hinge_bolts" displacement components 2 z
114 solid_mechanics_definitions boundary_conditions "pull_door_hinge_bolts" displacement scale_factor [0 0 9]
115 solid_mechanics_definitions boundary_conditions "pull_door_hinge_bolts" quadrature_options tessellation_option "tessellation_options_1"
116 #!=BOUNDARY CONDITIONS
117 #=INTERACTIONS
118 solid_mechanics_definitions interaction_properties "friction" new
119 solid_mechanics_definitions interaction_properties "friction" coulomb_friction coefficient_with_regularization friction_coefficient 0.3
120 solid_mechanics_definitions interaction_properties "friction" coulomb_friction coefficient_with_regularization regularization_velocity 0.1
121 solid_mechanics_definitions interactions "frictional_contact" new
122 solid_mechanics_definitions interactions "frictional_contact" mechanical_contact base_contact complement true
123 solid_mechanics_definitions interactions "frictional_contact" mechanical_contact interaction_properties "friction"
124 solid_mechanics_definitions interactions "frictional_contact" mechanical_contact approximate_tangent_iterations 6

```

```

125 solid_mechanics_definitions interactions "frictional_contact" quadrature_options tessellation_option "tessellation_options_1"
126 solid_mechanics_definitions interactions "frictional_contact" mechanical_contact activation_distance value 0.5
127 solid_mechanics_definitions interactions "frictional_contact" mechanical_contact activation_zone_influential_radius_factor 1.0
128 solid_mechanics_definitions interactions "frictional_contact" mechanical_contact potential_cutoff_distance_factor 0.5
129 #!=INTERACTIONS
130 #=-PROBES
131 solid_mechanics_definitions probes "door_link_reaction_force" new
132 solid_mechanics_definitions probes "door_link_reaction_force" integrated_surface_quantity
133 solid_mechanics_definitions probes "door_link_reaction_force" integrated_surface_quantity use_set_from_boundary_condition "hold_door_link_bottom"
134 solid_mechanics_definitions probes "door_link_reaction_force" integrated_surface_quantity variables reaction_force 0 x
135 solid_mechanics_definitions probes "door_link_reaction_force" integrated_surface_quantity variables reaction_force 1 y
136 solid_mechanics_definitions probes "door_link_reaction_force" integrated_surface_quantity variables reaction_force 2 z
137 solid_mechanics_definitions probes "door_hinge_reaction_force" new
138 solid_mechanics_definitions probes "door_hinge_reaction_force" integrated_surface_quantity
139 solid_mechanics_definitions probes "door_hinge_reaction_force" integrated_surface_quantity use_set_from_boundary_condition "pull_door_hinge_bolts"
140 solid_mechanics_definitions probes "door_hinge_reaction_force" integrated_surface_quantity variables reaction_force 0 x
141 solid_mechanics_definitions probes "door_hinge_reaction_force" integrated_surface_quantity variables reaction_force 1 y
142 solid_mechanics_definitions probes "door_hinge_reaction_force" integrated_surface_quantity variables reaction_force 2 z
143 solid_mechanics_definitions probes "door_hinge_max_vm" new
144 solid_mechanics_definitions probes door_hinge_max_vm field extremum part active
145 solid_mechanics_definitions probes door_hinge_max_vm field extremum part ""
146 solid_mechanics_definitions probes door_hinge_max_vm field extremum part ""
147 solid_mechanics_definitions probes door_hinge_max_vm field extremum part "door_hinge"
148 solid_mechanics_definitions probes door_hinge_max_vm field extremum maximum active
149 solid_mechanics_definitions probes door_hinge_max_vm field extremum maximum stress active
150 solid_mechanics_definitions probes door_hinge_max_vm field extremum maximum stress von_mises
151 solid_mechanics_definitions probes door_hinge_max_vm field variables stress new
152 solid_mechanics_definitions probes door_hinge_max_vm field variables stress 0 von_mises
153 #!=PROBES
154 #=-FIELD OUTPUT
155 solid_mechanics_definitions outputs "door_link_field_results" new
156 solid_mechanics_definitions outputs "door_link_field_results" field database_name "door_link_results"
157 solid_mechanics_definitions outputs "door_link_field_results" field interval "simulation_output_interval"
158 solid_mechanics_definitions outputs "door_link_field_results" field part "door_link"
159 solid_mechanics_definitions outputs "door_link_field_results" field variables displacement 0 x
160 solid_mechanics_definitions outputs "door_link_field_results" field variables displacement 1 y
161 solid_mechanics_definitions outputs "door_link_field_results" field variables displacement 2 z
162 solid_mechanics_definitions outputs "door_link_field_results" field variables stress 0 von_mises
163 solid_mechanics_definitions outputs "door_link_field_results" field variables plastic_strain 0 equivalent
164 solid_mechanics_definitions outputs "door_hinge_field_results" new
165 solid_mechanics_definitions outputs "door_hinge_field_results" field database_name "door_hinge_results"
166 solid_mechanics_definitions outputs "door_hinge_field_results" field interval "simulation_output_interval"
167 solid_mechanics_definitions outputs "door_hinge_field_results" field part "door_hinge"
168 solid_mechanics_definitions outputs "door_hinge_field_results" field variables displacement 0 x
169 solid_mechanics_definitions outputs "door_hinge_field_results" field variables displacement 1 y
170 solid_mechanics_definitions outputs "door_hinge_field_results" field variables displacement 2 z
171 solid_mechanics_definitions outputs "door_hinge_field_results" field variables stress 0 von_mises
172 solid_mechanics_definitions outputs "door_hinge_field_results" field variables plastic_strain 0 equivalent
173 solid_mechanics_definitions outputs "door_pin_field_results" new
174 solid_mechanics_definitions outputs "door_pin_field_results" field database_name "door_pin_results"
175 solid_mechanics_definitions outputs "door_pin_field_results" field interval "simulation_output_interval"
176 solid_mechanics_definitions outputs "door_pin_field_results" field part "door_pin"
177 solid_mechanics_definitions outputs "door_pin_field_results" field variables displacement 0 x
178 solid_mechanics_definitions outputs "door_pin_field_results" field variables displacement 1 y
179 solid_mechanics_definitions outputs "door_pin_field_results" field variables displacement 2 z
180 solid_mechanics_definitions outputs "door_pin_field_results" field variables stress 0 von_mises
181 solid_mechanics_definitions outputs "door_pin_field_results" field variables plastic_strain 0 equivalent
182 solid_mechanics_definitions outputs "door_link_reaction_force_results" new
183 solid_mechanics_definitions outputs "door_link_reaction_force_results" history interval "simulation_output_interval"
184 solid_mechanics_definitions outputs "door_link_reaction_force_results" history probe_variables 0 door_link_reaction_force
185 solid_mechanics_definitions outputs "door_hinge_reaction_force_results" new
186 solid_mechanics_definitions outputs "door_hinge_reaction_force_results" history interval "simulation_output_interval"
187 solid_mechanics_definitions outputs "door_hinge_reaction_force_results" history probe_variables 0 door_hinge_reaction_force
188 solid_mechanics_definitions outputs "door_hinge_max_vm_results" new
189 solid_mechanics_definitions outputs "door_hinge_max_vm_results" history interval "simulation_output_interval"
190 solid_mechanics_definitions outputs "door_hinge_max_vm_results" history probe_variables 0 door_hinge_max_vm
191 #!=FIELD OUTPUT
192 #=-PROCEDURES
193 label "door_hinge_failure"
194 procedures "pull_door_hinge" solid_mechanics
195 procedures "pull_door_hinge" solid_mechanics interval "simulation_interval"
196 procedures "pull_door_hinge" solid_mechanics flex_model "flex_models_1"
197 procedures "pull_door_hinge" solid_mechanics time_stepping_method "implicit_dynamics_quasistatic"
198 procedures "pull_door_hinge" solid_mechanics boundary_conditions 0 "hold_door_link_bottom"
199 procedures "pull_door_hinge" solid_mechanics boundary_conditions 1 "pull_door_hinge_bolts"
200 procedures "pull_door_hinge" solid_mechanics interactions 0 "frictional_contact"
201 procedures "pull_door_hinge" solid_mechanics outputs 0 "door_link_field_results"
202 procedures "pull_door_hinge" solid_mechanics outputs 1 "door_hinge_field_results"
203 procedures "pull_door_hinge" solid_mechanics outputs 2 "door_pin_field_results"
204 procedures "pull_door_hinge" solid_mechanics outputs 3 "door_link_reaction_force_results"
205 procedures "pull_door_hinge" solid_mechanics outputs 4 "door_hinge_reaction_force_results"
206 procedures "pull_door_hinge" solid_mechanics outputs 5 door_hinge_max_vm_results
207 #!=PROCEDURES
208 #=-Save the Coreform Flex project
209 save "door_lock_assembly.cf"
210 #!=Save the Coreform Flex project
211 #=-SUBMIT TRIM JOB
212 # Check the trimmed mesh
213 job "job_door_hinge_trim" new
214 job "job_door_hinge_trim" trim processor_count 16
215 job "job_door_hinge_trim" trim trim_parts [door_link door_hinge door_pin]
216 job "job_door_hinge_trim" submit

```

```

217 job "job_door_hinge_trim" wait
218 #!=SUBMIT TRIM JOB
219 #=SUBMIT SIMULATION JOB
220 job "job_door_hinge_failure" new
221 job "job_door_hinge_failure" simulation processor_count 16
222 job "job_door_hinge_failure" submit
223 job "job_door_hinge_failure" wait
224 #!=SUBMIT SIMULATION JOB
225 #=VISUALIZE SIMULATION RESULTS
226 job "job_door_hinge_failure" render_iga timestep last scalars stress von_mises
227 #!=VISUALIZE SIMULATION RESULTS
228

```

4.2.2. door_hinge/job_door_hinge_failure.iga

```

1 {
2   "coreform_iga_version": "2025.8",
3   "data_output_file": "cf_iga_data_output_ip2.json",
4   "flex_models": [
5     {
6       "database_name": "geom",
7       "label": "flex_models_1",
8       "parts": [
9         {
10          "label": "door_hinge_instance",
11          "material": "aluminum",
12          "material_model": "neohookean_isotropic_plasticity",
13          "part": "door_hinge",
14          "quadrature_options": {
15            "optimization_lawson_hanson_solve_completion": true,
16            "optimization_method": "tet_pruning",
17            "optimization_residual_tolerance": 1.0000000000000002e-12,
18            "optimized_cut_quadrature_strategy": "reduce_from_uncut",
19            "quadrature": "QP1",
20            "tessellation_option": "tessellation_options_1"
21          }
22        },
23        {
24          "label": "door_link_instance",
25          "material": "aluminum",
26          "material_model": "neohookean_isotropic_plasticity",
27          "part": "door_link",
28          "quadrature_options": {
29            "optimization_lawson_hanson_solve_completion": true,
30            "optimization_method": "tet_pruning",
31            "optimization_residual_tolerance": 1.0000000000000002e-12,
32            "optimized_cut_quadrature_strategy": "reduce_from_uncut",
33            "quadrature": "QP1",
34            "tessellation_option": "tessellation_options_1"
35          }
36        },
37        {
38          "label": "door_pin_instance",
39          "material": "steel",
40          "material_model": "neohookean_isotropic_plasticity",
41          "part": "door_pin",
42          "quadrature_options": {
43            "optimization_lawson_hanson_solve_completion": true,
44            "optimization_method": "tet_pruning",
45            "optimization_residual_tolerance": 1.0000000000000002e-12,
46            "optimized_cut_quadrature_strategy": "reduce_from_uncut",
47            "quadrature": "QP1",
48            "tessellation_option": "tessellation_options_1"
49          }
50        }
51      ],
52      "small_cell_volume_ratio": 0
53    }
54  ],
55  "functions": [
56    {
57      "label": "aluminum_plastic_hardening",
58      "piecewise_linear": {
59        "abscissa": [0, 10],
60        "ordinate": [0, 100]
61      }
62    },
63    {
64      "label": "steel_plastic_hardening",
65      "piecewise_linear": {
66        "abscissa": [0, 10],
67        "ordinate": [0, 100]
68      }
69    }
70  ]
71 }

```

```

70 {
71   "constant": {
72     "value": 1
73   },
74   "label": "constant_1"
75 },
76 {
77   "label": "linear_ramp",
78   "piecewise_linear": {
79     "abscissa": [0, 1],
80     "ordinate": [0, 1]
81   }
82 },
83 ],
84 "intervals": [
85   {
86     "label": "simulation_interval",
87     "start_time": 0,
88     "stop_time": 1,
89     "time_increment": 0.0005
90   },
91   {
92     "label": "simulation_output_interval",
93     "step_increment": 1,
94     "time_increment": 1,
95     "use_start_stop_from_interval": "simulation_interval"
96   }
97 ],
98 "label": "door_hinge_failure",
99 "linear_equation_solvers": [
100   {
101     "direct": {
102       "lu": {
103         "use_diagonal_scaling": false
104       }
105     },
106     "label": "direct_lu"
107   },
108   {
109     "direct": {
110       "multi_frontal": {
111         "use_diagonal_scaling": false
112       }
113     },
114     "label": "direct_multi_frontal"
115   }
116 ],
117 "materials": [
118   {
119     "label": "aluminum",
120     "mass_density": 2.6999999999999998e-9,
121     "neohookean_isotropic_plasticity": {
122       "isotropic_plasticity": {
123         "hardening_function": "aluminum_plastic_hardening",
124         "yield_stress": 322
125       },
126       "neohookean": {
127         "poissons_ratio": 0.3,
128         "youngs_modulus": 73800
129       }
130     },
131   },
132   {
133     "label": "steel",
134     "mass_density": 2.6999999999999998e-9,
135     "neohookean_isotropic_plasticity": {
136       "isotropic_plasticity": {
137         "hardening_function": "steel_plastic_hardening",
138         "yield_stress": 2000
139       },
140       "neohookean": {
141         "poissons_ratio": 0.29,
142         "youngs_modulus": 200000
143       }
144     },
145   }
146 ],
147 "nonlinear_equation_solvers": [
148   {
149     "label": "newton_raphson",
150     "newton": {
151       "linear_equation_solver": "direct_multi_frontal",
152       "maximum_iterations": 20,
153       "target_relative_residual": 1e-8,
154       "use_line_search": false
155     }
156   },
157 ],
158 "procedures": [
159   {
160     "label": "pull_door_hinge",
161     "solid_mechanics": {

```

```

162     "boundary_conditions": [
163         "hold_door_link_bottom",
164         "pull_door_hinge_bolts"
165     ],
166     "flex_model": "flex_models_1",
167     "interactions": [
168         "frictional_contact"
169     ],
170     "interval": "simulation_interval",
171     "outputs": [
172         "door_link_field_results",
173         "door_hinge_field_results",
174         "door_pin_field_results",
175         "door_link_reaction_force_results",
176         "door_hinge_reaction_force_results"
177     ],
178     "time_stepping_method": "implicit_dynamics_quasistatic"
179 }
180 ],
181 ],
182 "solid_mechanics_definitions": {
183     "boundary_conditions": [
184         {
185             "displacement": {
186                 "components": ["x", "y", "z"],
187                 "function": "constant_1",
188                 "scale_factor": 0
189             },
190             "label": "hold_door_link_bottom",
191             "penalty": {
192                 "dimensionless": true,
193                 "value": 10
194             },
195             "quadrature_options": {
196                 "optimization_lawson_hanson_solve_completion": true,
197                 "optimization_method": "tet_pruning",
198                 "optimization_residual_tolerance": 1.0000000000000002e-12,
199                 "optimized_cut_quadrature_strategy": "reduce_from_uncut",
200                 "quadrature": "QP1",
201                 "tessellation_option": "tessellation_options_1"
202             },
203             "set": "door_link_bottom_surfaces"
204         },
205         {
206             "displacement": {
207                 "components": ["x", "y", "z"],
208                 "function": "linear_ramp",
209                 "scale_factor": [0, 0, 9]
210             },
211             "label": "pull_door_hinge_bolts",
212             "penalty": {
213                 "dimensionless": true,
214                 "value": 10
215             },
216             "quadrature_options": {
217                 "optimization_lawson_hanson_solve_completion": true,
218                 "optimization_method": "tet_pruning",
219                 "optimization_residual_tolerance": 1.0000000000000002e-12,
220                 "optimized_cut_quadrature_strategy": "reduce_from_uncut",
221                 "quadrature": "QP1",
222                 "tessellation_option": "tessellation_options_1"
223             },
224             "set": "door_hinge_bolt_surfaces"
225         }
226     ],
227     "interaction_properties": [
228         {
229             "coulomb_friction": {
230                 "coefficient_with_regularization": {
231                     "friction_coefficient": 0.3,
232                     "regularization_velocity": 0.1
233                 }
234             },
235             "label": "friction"
236         }
237     ],
238     "interactions": [
239         {
240             "label": "frictional_contact",
241             "mechanical_contact": {
242                 "activation_distance": {
243                     "dimensionless": true,
244                     "value": 0.6
245                 },
246                 "activation_zone_influential_radius_factor": 1,
247                 "angle_scaling": {
248                     "negative_dot_product_lower_limit": 0,
249                     "negative_dot_product_upper_limit": 1,
250                     "ramp_function_continuity": "C0",
251                     "scaling_strategy": "signed_distance"
252                 },
253                 "approximate_tangent_iterations": 6,

```

```

254     "base_contact": {
255         "complement": true
256     },
257     "collision_search_filter": {
258         "angle_filter": {
259             "threshold_angle": 30
260         },
261         "area_filter": {
262             "threshold_area_ratio": 10
263         },
264         "spurious_self_contact_filter": {
265             "use": true
266         }
267     },
268     "effective_distance_smoothing": {
269         "distance_smoothing_factor": 0.125
270     },
271     "interaction_properties": "friction",
272     "penalty": {
273         "dimensionless": true,
274         "value": 10
275     },
276     "potential_cutoff_distance_factor": 0.5
277 },
278 "quadrature_options": {
279     "optimization_lawson_hanson_solve_completion": true,
280     "optimization_method": "tet_pruning",
281     "optimization_residual_tolerance": 1.0000000000000002e-12,
282     "optimized_cut_quadrature_strategy": "reduce_from_uncut",
283     "quadrature": "QP1",
284     "tessellation_option": "tessellation_options_1"
285 }
286 ],
287 "outputs": [
288     {
289         "field": {
290             "database_name": "door_link_results_0p6",
291             "element_variable_output_strategy": "interpolate",
292             "external_surfaces_only": false,
293             "include_integration_point_output": false,
294             "interval": "simulation_output_interval",
295             "part": "door_link",
296             "variables": {
297                 "displacement": ["x", "y", "z"],
298                 "plastic_strain": ["equivalent"],
299                 "stress": ["von_mises"]
300             }
301         },
302         "label": "door_link_field_results"
303     },
304     {
305         "field": {
306             "database_name": "door_hinge_results_0p6",
307             "element_variable_output_strategy": "interpolate",
308             "external_surfaces_only": false,
309             "include_integration_point_output": false,
310             "interval": "simulation_output_interval",
311             "part": "door_hinge",
312             "variables": {
313                 "displacement": ["x", "y", "z"],
314                 "plastic_strain": ["equivalent"],
315                 "stress": ["von_mises"]
316             }
317         },
318         "label": "door_hinge_field_results"
319     },
320     {
321         "field": {
322             "database_name": "door_pin_results_0p6",
323             "element_variable_output_strategy": "interpolate",
324             "external_surfaces_only": false,
325             "include_integration_point_output": false,
326             "interval": "simulation_output_interval",
327             "part": "door_pin",
328             "variables": {
329                 "displacement": ["x", "y", "z"],
330                 "plastic_strain": ["equivalent"],
331                 "stress": ["von_mises"]
332             }
333         },
334         "label": "door_pin_field_results"
335     },
336     {
337         "history": {
338             "interval": "simulation_output_interval",
339             "probe_variables": ["door_link_reaction_force"]
340         },
341         "label": "door_link_reaction_force_results"
342     },
343     {
344         "history": {
345

```

```

346         "interval": "simulation_output_interval",
347         "probe_variables": ["door_hinge_reaction_force"]
348     },
349     "label": "door_hinge_reaction_force_results"
350 },
351 ],
352 "probes": [
353     {
354         "integrated_surface_quantity": {
355             "use_set_from_boundary_condition": "hold_door_link_bottom",
356             "variables": {
357                 "reaction_force": ["x", "y", "z"]
358             }
359         },
360         "label": "door_link_reaction_force"
361     },
362     {
363         "integrated_surface_quantity": {
364             "use_set_from_boundary_condition": "pull_door_hinge_bolts",
365             "variables": {
366                 "reaction_force": ["x", "y", "z"]
367             }
368         },
369         "label": "door_hinge_reaction_force"
370     }
371 ],
372 },
373 "tessellation_options": [
374     {
375         "label": "tessellation_options_1",
376         "meshgems": {
377             "average_triangle_edge_subdivisions": 4
378         }
379     }
380 ],
381 "time_steppers": [
382     {
383         "generalized_alpha": {
384             "adaptivity": {
385                 "decrease_factor": 0.25,
386                 "increase_factor": 1.5,
387                 "maximum_time_step": 0.1,
388                 "minimum_time_step": 0.00001
389             },
390             "nonlinear_equation_solver": "newton_raphson",
391             "predictor": "constant_velocity",
392             "spectral_radius": 0
393         },
394         "label": "implicit_dynamics_quasistatic"
395     }
396 ]
397 }

```

4.2.3. door_hinge/read_force_vm.py

```

1  #!/usr/bin/env python3
2  # -*- coding: utf-8 -*-
3  """
4  Created on Thu Aug 21 10:42:45 2025
5
6  @author: danajer
7  """
8
9  import pandas as pd
10 import matplotlib.pyplot as plt
11 import numpy as np
12 import seaborn as sns
13
14 plt.close('all')
15
16 from matplotlib import rc
17
18 def set_style_sns():
19     sns.set_context('paper')
20     sns.set(font = 'serif')
21     sns.set(font_scale = 1.1)
22
23     sns.set_style('white', {
24         'font.family': 'serif',
25         'font.serif': ['Time', 'Palatino', 'serif'],
26         'lines.markersize': 10
27     })
28
29

```

```

30 plt.rcParams.update({'font.size':8})
31 set_style('sns')
32
33 plt.rcParams.update({'text.usetex': False})
34
35 files = ['with_probe/results_0p8/cf_iga_data_output.json',
36         'with_probe/results_1p0/cf_iga_data_output.json',
37         'with_probe/results_1p2/cf_iga_data_output.json']
38 mesh_sizes = ['0p8', '1p0', '1p2']
39 directs = ['x', 'y', 'z']
40
41 plt.figure(figsize = (10,8))
42
43 force_mag = []
44 vm_stresses = []
45 dofs = np.array([260247, 154746, 101739])
46 runtimes = []
47 for ii,file in enumerate(files):
48     data = pd.read_json(file, orient = 'index')
49     data_f = data['history']['pull_door_hinge']['door_hinge_reaction_force']
50     time = data_f['time']
51     force = np.stack([data_f['reaction_force']['x'], data_f['reaction_force']['y'], data_f['reaction_force']['z']])
52     vm_stress = data['history']['pull_door_hinge']['door_hinge_max_vm']['extremum']['stress']['von_mises']
53     vm_stresses.append(vm_stress[-1])
54     force_mag.append(np.sqrt(force[0,-1]**2 + force[1,-1]**2 + force[2,-1]**2))
55     runtimes.append(data['monitors']['pull_door_hinge']['timing']['procedure_solve'][0])
56     for jj in range(3):
57         plt.subplot(2,2,jj+1)
58         plt.plot(time, force[jj,:], label = mesh_sizes[ii])
59
60
61         if ii ==2:
62             plt.xlabel('Time, s')
63             plt.ylabel('door hinge reaction force, lbf')
64             plt.title(directs[jj])
65             plt.legend()
66             plt.grid('on')
67
68     plt.subplot(2,2,4)
69     plt.plot(time, vm_stress, label = mesh_sizes[ii])
70     if ii ==2:
71         plt.xlabel('Time, s')
72         plt.ylabel('door hinge von Mises, psi')
73         plt.title(directs[jj])
74         plt.legend()
75         plt.grid('on')
76
77
78 force_mag = np.array(force_mag)
79 vm_stresses = np.array(vm_stresses)
80 runtimes = np.array(runtimes)
81
82 plt.tight_layout()
83 plt.savefig('reaction_force_comparison.png')
84
85 plt.show()
86
87
88 plt.figure(figsize = (8,8))
89 plt.subplot(2,1,1)
90 plt.plot(dofs, force_mag, '-.', label = 'IGA')
91 plt.ylabel('Reaction Force Magnitude, lbf')
92 plt.xlabel('Number of DOFs')
93 plt.grid('on')
94 plt.legend()
95 plt.tight_layout()
96
97 plt.subplot(2,1,2)
98 plt.plot(dofs, vm_stresses, '-.', label = 'IGA')
99 plt.ylabel('Peak von Mises, psi')
100 plt.xlabel('Number of DOFs')
101 plt.grid('on')
102 plt.legend()
103 plt.tight_layout()
104 plt.savefig('door_hinge_force_vm_dofs.png')
105 plt.show()

```

5. METHODS

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6. RESULTS

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7. CONCLUSION

That is, as they say, all, folks.

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APPENDIX A. Appendices

Include any supplementary material, such as additional data, calculations, or detailed explanations of methods that are too lengthy for the main sections.



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